IMAT 모의고사 10회 (고난도)

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> When a new drug is approved, its benefits are usually reported as a relative risk reduction: "cuts the risk of stroke by 50%". This figure can sound impressive even when the absolute benefit is small. If a stroke would occur in 2 of every 100 untreated patients and in 1 of every 100 treated patients, the relative reduction is 50%, but the absolute reduction is only one percentage point — meaning 100 patients must be treated for one to benefit. Communicating both numbers, some argue, would allow patients to weigh benefits against side effects more realistically.

1. Based on the passage, if a drug reduces the risk of an event from 4% to 3%, what is the "number needed to treat" for one patient to benefit?

A. 1
B. 100
C. 75
D. 25
E. 4

2. The author implies that reporting only relative risk reduction may:

A. be illegal.
B. understate the benefits of drugs.
C. be the most accurate way to describe a drug.
D. make the benefits of a drug appear larger than they are in practical terms.
E. increase side effects.

3. Which scientist formulated the laws of inheritance by studying pea plants in the 19th century?

A. Francis Crick
B. Thomas Hunt Morgan
C. Gregor Mendel
D. Hugo de Vries
E. Charles Darwin

4. The Hippocratic Oath is traditionally associated with which ancient civilisation?

A. Persian
B. Roman
C. Egyptian
D. Chinese
E. Greek

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "No drug that causes drowsiness is allowed for pilots on duty. Some antihistamines cause drowsiness." Which conclusion follows?

A. No antihistamines are allowed for pilots on duty.
B. Some drugs that cause drowsiness are allowed.
C. All drugs not allowed for pilots cause drowsiness.
D. All antihistamines are allowed for pilots on duty.
E. Some antihistamines are not allowed for pilots on duty.

6. Solution A has a concentration of 10% and solution B of 30%. In what ratio (A : B) must they be mixed to obtain a 15% solution?

A. 3 : 2
B. 1 : 3
C. 2 : 1
D. 3 : 1
E. 1 : 1

7. A clock shows 3:00. What is the smaller angle between the hour and minute hands at 3:30?

A. 120°
B. 90°
C. 75°
D. 60°
E. 105°

8. Five numbers have a mean of 10 and a median of 8. If the largest number is increased by 10, what are the new mean and median?

A. Mean 10, median 10
B. Mean 20, median 18
C. Mean 12, median 10
D. Mean 10, median 8
E. Mean 12, median 8

9. Water evaporates from a pool at a rate of 2% of the current volume per day. Approximately what fraction of the original volume remains after 2 days?

A. 0.9216
B. 0.96
C. 0.98
D. 0.9604
E. 0.94

Section 3 — Biology (Q10–32)

10. A protein is found to have a signal sequence and ends up in the lumen of the lysosome. Which tag is responsible for sorting it to lysosomes in the Golgi?

A. Ubiquitin
B. Nuclear localisation signal
C. KDEL sequence
D. Mannose-6-phosphate
E. Poly-A tail

11. Telomerase is particularly active in which of the following?

A. Mature neurons
B. Platelets
C. Red blood cells
D. Germ cells and most cancer cells
E. Muscle fibres

12. A drug blocks the formation of the mitotic spindle. In which phase will cells accumulate?

A. Metaphase (M phase)
B. G2
C. G0
D. G1
E. S

13. Cyanide inhibits cytochrome c oxidase (complex IV). What is the immediate effect?

A. The Krebs cycle speeds up.
B. Glycolysis stops completely.
C. Oxygen consumption increases.
D. Electron transport and oxidative ATP production stop.
E. The proton gradient increases.

14. A man affected by an X-linked recessive disorder marries a carrier woman. What is the probability that a daughter is affected?

A. 1
B. 1/2
C. 0
D. 3/4
E. 1/4

15. In C_4 plants, CO_2 is initially fixed into:

A. a 4-carbon compound by PEP carboxylase.
B. glucose directly.
C. oxygen.
D. a 3-carbon compound by RuBisCO.
E. ATP.

16. Which of the following best explains why carbon monoxide is dangerous?

A. It blocks the trachea.
B. It destroys red blood cells.
C. It dissolves in plasma and lowers pH.
D. It increases ventilation rate.
E. It binds haemoglobin with much higher affinity than O_2, reducing oxygen transport.

17. Which molecule is the immediate energy source for muscle contraction?

A. Fatty acids
B. ATP
C. Creatine
D. Glycogen
E. Glucose

18. A drug blocks voltage-gated K^+ channels in neurons. Which effect is expected?

A. Faster repolarisation
B. No depolarisation
C. Hyperpolarisation at rest
D. Increased myelination
E. Prolonged action potentials due to slowed repolarisation

19. Starling forces at the arterial end of a capillary favour:

A. filtration of fluid out of the capillary due to higher hydrostatic pressure.
B. movement of red blood cells into tissues.
C. no fluid movement.
D. movement of proteins out of the capillary.
E. reabsorption of fluid due to high oncotic pressure.

20. Two genes are 20 map units apart. A double heterozygote AB/ab is test-crossed. What proportion of offspring will be Ab?

A. 80%
B. 20%
C. 40%
D. 5%
E. 10%

21. Which of the following is correct about hormones derived from cholesterol?

A. They are stored in secretory granules for long periods.
B. They bind to receptors on the cell surface only.
C. They are peptides.
D. They can cross the plasma membrane and often bind intracellular receptors.
E. They are water-soluble and travel freely in plasma.

22. A DNA sample melts (denatures) at a higher temperature than another. The first sample most likely has:

A. a higher proportion of A–T base pairs.
B. a higher proportion of G–C base pairs.
C. single-stranded regions only.
D. more uracil.
E. fewer phosphate groups.

23. Rhesus disease of the newborn can occur when:
A. both parents are Rh^-.
B. the mother has blood group AB.
C. the fetus is Rh^- and the father is Rh^-.
D. an Rh^+ mother carries an Rh^- fetus.
E. an Rh^- mother, previously sensitised, carries an Rh^+ fetus.

24. A loop diuretic inhibits the Na^+–K^+–$2Cl^-$ cotransporter. Where does it act?
A. Descending limb
B. Glomerulus
C. Collecting duct
D. Thick ascending limb of the loop of Henle
E. Proximal convoluted tubule

25. A cell undergoing meiosis has 4 chromosomes in G1 (2n = 4). How many DNA molecules (chromatids) are present in each cell at metaphase II?
A. 8
B. 1
C. 4
D. 16
E. 2

26. In an enzyme-catalysed reaction, a non-competitive inhibitor:
A. has no effect on the reaction rate.
B. decreases Vmax without changing Km.
C. binds only to the active site.
D. increases Vmax.
E. increases Km without changing Vmax.

27. A mutation in the promoter region of a gene is most likely to affect:
A. the rate of transcription of the gene.
B. tRNA charging.
C. the stop codon.
D. splicing of introns.
E. the amino acid sequence of the protein.

28. Which of the following explains how glucose is absorbed into intestinal epithelial cells against its concentration gradient?
A. Secondary active transport coupled to Na^+ (SGLT1)
B. Osmosis
C. Endocytosis
D. Primary active transport using ATP directly on glucose
E. Simple diffusion

29. Penicillin is ineffective against Mycoplasma bacteria because they:
A. are viruses.
B. lack ribosomes.
C. are too large.
D. have a nucleus.
E. lack a peptidoglycan cell wall.

30. Which hormone, released from the duodenum in response to fats, stimulates gallbladder contraction?

A. Somatostatin
B. Secretin
C. Insulin
D. Gastrin
E. Cholecystokinin (CCK)

31. Alternative splicing allows:

A. tRNA to recognise several codons.
B. prokaryotes to remove introns.
C. one gene to code for multiple different proteins.
D. DNA to replicate faster.
E. ribosomes to read mRNA in reverse.

32. In a population of 1000 individuals in Hardy–Weinberg equilibrium, 90 show the recessive phenotype. How many individuals are expected to be heterozygous?

A. 490
B. 210
C. 700
D. 420
E. 90

Section 4 — Chemistry (Q33–47)

33. A hydrocarbon contains 85.7% carbon by mass and has a molar mass of 56 g/mol. What is its molecular formula? (C = 12, H = 1)

A. C_3H_6
B. C_4H_8
C. C_2H_4
D. C_5H_{12}
E. C_4H_{10}

34. Balance in acidic solution: $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$. How many H^+ ions appear on the reactant side?

A. 7
B. 8
C. 16
D. 14
E. 6

35. 25 mL of 0.1 mol/L NaOH is titrated with 0.1 mol/L HCl. After 24.9 mL of HCl has been added, the solution is:

A. strongly acidic.
B. a buffer.
C. strongly acidic with excess HCl.
D. exactly neutral.
E. slightly basic.

36. Which of the following ions has the same electron configuration as argon?
A. Mg^{2+}
B. K^{+}
C. F^{-}
D. Li^{+}
E. Na^{+}

37. What is the hybridisation of the carbon atoms in ethyne (C_2H_2)?
A. sp^2
B. sp
C. sp^3d
D. sp^3
E. sp^3d^2

38. 2.4 g of magnesium (Mg = 24) reacts with excess HCl: $Mg + 2HCl \rightarrow MgCl_2 + H_2$. What volume of H_2 is produced at STP (22.4 L/mol)?
A. 1.12 L
B. 2.24 L
C. 0.224 L
D. 4.48 L
E. 22.4 L

39. Which of the following alcohols cannot be oxidised by acidified potassium dichromate?
A. Methanol
B. Ethanol
C. 2-methylpropan-2-ol
D. Propan-2-ol
E. Propan-1-ol

40. What mass of NaOH (M = 40 g/mol) is needed to prepare 200 mL of a solution with pH 13 at 25 °C?
A. 4 g
B. 0.4 g
C. 0.8 g
D. 0.08 g
E. 8 g

41. Which of the following has the highest lattice energy?
A. KCl
B. MgO
C. KBr
D. NaCl
E. NaBr

42. The pKa of acetic acid is 4.76. A buffer contains 0.1 mol/L acetic acid and 1.0 mol/L sodium acetate. What is its pH?
A. 3.76
B. 4.76
C. 5.76
D. 6.76
E. 7.00

43. For the reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$, Kc = 4 at a given temperature. What is Kc for $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ at the same temperature?

A. 4
B. −4
C. 0.25
D. 16
E. 2

44. How many stereoisomers does a molecule with two different chiral centres (and no internal symmetry) have at most?

A. 4
B. 1
C. 8
D. 3
E. 2

45. An element X forms an oxide X_2O_3. Which group is X most likely in?

A. Group 2
B. Group 1
C. Group 13
D. Group 17
E. Group 15

46. In a concentration cell, both electrodes are copper in $CuSO_4$ solutions of 1 mol/L and 0.01 mol/L. Electrons flow in the external circuit:

A. from the 0.01 mol/L electrode to the 1 mol/L electrode.
B. from the 1 mol/L electrode to the 0.01 mol/L electrode.
C. only if a salt bridge is absent.
D. in both directions equally.
E. not at all.

47. Hydrolysis of a triglyceride with NaOH (saponification) produces:

A. esters and water.
B. glucose and fatty acids.
C. glycerol and fatty acid sodium salts (soap).
D. ethanol and acetic acid.
E. amino acids.

Section 5 — Physics & Mathematics (Q48–60)

48. A ball is thrown at 20 m/s at 30° above the horizontal from ground level. What is its time of flight? ($g = 10\ m/s^2$)

A. 1 s
B. 1.73 s
C. 3.46 s
D. 2 s
E. 4 s

49. A 4 kg box is pulled across a rough floor at constant velocity by a horizontal force of 12 N. What is the coefficient of kinetic friction? ($g = 10\ m/s^2$)

A. 0.3
B. 3
C. 0.4
D. 0.03
E. 0.12

50. A 9 V battery with internal resistance 1 Ω is connected to an external 2 Ω resistor. What is the terminal voltage?

A. 6 V
B. 3 V
C. 4.5 V
D. 9 V
E. 7.5 V

51. The intensity of a sound increases from $10^{-8}\ W/m^2$ to $10^{-6}\ W/m^2$. By how many decibels does the sound level increase?

A. 100 dB
B. 2 dB
C. 10 dB
D. 200 dB
E. 20 dB

52. According to Poiseuille's law, if the radius of a blood vessel is halved (other factors constant), the flow rate becomes:

A. 1/2 of the original.
B. 1/4 of the original.
C. 1/16 of the original.
D. 1/8 of the original.
E. 2 times the original.

53. A satellite orbits Earth in a circular orbit. If its orbital radius were doubled, the gravitational force on it would become:

A. one quarter as large.
B. unchanged.
C. 4 times larger.
D. 2 times larger.
E. half as large.

54. A 0.2 kg sample of metal at 100 °C is placed in 0.4 kg of water at 20 °C. The final temperature is 30 °C. Ignoring losses, what is the specific heat capacity of the metal? ($c_{water} = 4200\ J/(kg\cdot°C)$)

A. 1200 J/(kg·°C)
B. 600 J/(kg·°C)
C. 12 000 J/(kg·°C)
D. 2400 J/(kg·°C)
E. 4200 J/(kg·°C)

55. Solve for x: $\log_3(x + 1) - \log_3(x - 1) = 1$.

A. 4
B. 3
C. 1
D. 2
E. No solution

56. How many solutions does the equation $2 \sin x = 1$ have in the interval $[0, 2\pi)$?

A. 1
B. 0
C. 2
D. 3
E. 4

57. A disease affects 1% of a population. A test detects it in 90% of affected people and gives a false positive in 9% of unaffected people. If a person tests positive, what is the approximate probability that they have the disease?

A. 1%
B. 50%
C. 10%
D. 90%
E. 9%

58. The area enclosed between $y = x^2$ and $y = 4$ is:

A. 64/3
B. 32/3
C. 8
D. 16
E. 16/3

59. For what value(s) of k does the equation $x^2 + kx + 9 = 0$ have exactly one real solution?

A. $k = \pm 6$
B. $k = 3$
C. $k = \pm 3$
D. $k = 9$
E. $k = 6$

60. A radioactive isotope has a decay constant giving a half-life of 8 days. A sample initially has an activity of 3200 Bq. After how many days is the activity 200 Bq?

A. 64 days
B. 32 days
C. 24 days
D. 40 days
E. 16 days

정답 및 해설

번호	정답	해설
1	B	절대위험감소 = 1%p → NNT = 1/0.01 = 100.
2	D	상대위험만 제시하면 실질적 이득이 과장되어 보임.
3	C	멘델.
4	E	고대 그리스.
5	E	졸음 유발 항히스타민제는 금지 → 일부 항히스타민제는 금지.
6	D	교차법: (30−15) : (15−10) = 15 : 5 = 3 : 1.
7	C	시침 105°(90+15), 분침 180° → 75°.
8	E	합 50 → 60, 평균 12. 최댓값 변화는 중앙값 불변 → 8.
9	D	$0.98^2 = 0.9604$.
10	D	만노스-6-인산 표지.
11	D	생식세포·대부분의 암세포.
12	A	방추체 형성 저해 → 중기에서 정지(방추체 점검점).
13	D	전자전달 정지 → 산화적 ATP 생성 중단.
14	B	$X^aY \times X^AX^a$ → 딸: X^aX^A 또는 X^aX^a → 1/2.
15	A	PEP 카복실레이스 → 4탄소 옥살로아세트산.
16	E	Hb 친화력 O_2의 약 200배 이상.
17	B	ATP.
18	E	재분극 지연 → 활동전위 연장.
19	A	동맥 끝: 정수압 > 교질삼투압 → 여과.
20	E	재조합 20% → Ab, aB 각 10%.
21	D	스테로이드는 막 통과, 세포 내 수용체.
22	B	G–C 비율 높을수록 Tm 높음.
23	E	감작된 Rh^- 산모 + Rh^+ 태아.
24	D	헨레 고리 두꺼운 상행지(푸로세마이드).
25	C	감수 2분열 중기: 염색체 2개 × 염색분체 2 = 4.
26	B	비경쟁적 저해: Vmax↓, Km 불변.
27	A	프로모터 → 전사 속도/여부.
28	A	SGLT1: Na^+ 기울기를 이용한 2차 능동수송.
29	E	마이코플라스마는 세포벽 없음.
30	E	CCK.
31	C	한 유전자 → 여러 단백질.
32	D	$q^2 = 0.09$ → q = 0.3, p = 0.7 → 2pq = 0.42 → 420명.
33	B	C:H = 7.14 : 14.3 = 1 : 2 → $CH_2(14) \times 4 = 56$ → C_4H_8.
34	D	$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$.
35	E	아직 NaOH 0.1 mL분 과량 → 약한 염기성.

번호	정답	해설
36	B	K^+ (18 e^-).
37	B	삼중결합 → sp.
38	B	0.1 mol Mg → 0.1 mol H_2 = 2.24 L.
39	C	3차 알코올은 산화 안 됨.
40	C	pOH 1 → $[OH^-]$ 0.1 × 0.2 L = 0.02 mol × 40 = 0.8 g.
41	B	MgO: 2+/2− 전하, 작은 이온 반경.
42	C	pH = 4.76 + log(1.0/0.1) = 5.76.
43	C	역반응 K = 1/4.
44	A	$2^n = 4$.
45	C	X^{3+} → 13족(예: Al_2O_3).
46	A	묽은 쪽(0.01)이 산화 전극(anode) → 전자는 묽은 쪽 → 진한 쪽.
47	C	글리세롤 + 지방산 나트륨염(비누).
48	D	수직 성분 10 m/s → t = 2×10/10 = 2 s.
49	A	12 = μ×40 → μ = 0.3.
50	A	I = 9/3 = 3 A → 단자전압 3×2 = 6 V.
51	E	$10 \log(10^2) = 20$ dB.
52	C	$Q \propto r^4$ → $(1/2)^4 = 1/16$.
53	A	$F \propto 1/r^2$.
54	A	물이 얻은 열 0.4×4200×10 = 16 800 J = 0.2 × c × 70 → c = 1200.
55	D	(x+1)/(x−1) = 3 → x = 2.
56	C	sin x = 1/2 → π/6, 5π/6.
57	E	양성: 0.009 + 0.0891 = 0.0981 → 0.009/0.0981 ≈ 9%.
58	B	$\int_{-2}^{2} (4 - x^2)\,dx = 16 - 16/3 = 32/3$.
59	A	판별식 $k^2 - 36 = 0$ → k = ±6.
60	B	3200 → 200 = 1/16 → 4 반감기 × 8 = 32일.